

Welcome to

DESIGNCON[®] 2024

WHERE THE CHIP MEETS THE BOARD

Conference

January 30 – February 1, 2024

Santa Clara Convention Center

Expo

January 31 – February 1, 2024



Are 1.0 mm Precision RF Connectors Really Required for 224 Gbps-PAM4 Verification?

Brandon Gore¹

Richard Mellitz¹, Andrew Josephson¹

Francesco de Paulis², Luis Boluna³, John Calvin³, Rick Rabinovich³, Mike Resso³

¹Samtec Inc, ²University of L'Aquila, ³Keysight Technologies



SPEAKERS



Brandon T. Gore, PhD

SI Technologist, Samtec Inc.

Brandon.gore@samtec.com

Brandon Gore is presently a Principal Technologist at Samtec managing both the Signal Integrity R&D and Electronic Industry Standards teams. His research focuses are advanced interconnect materials, glass packaging, direct drive optics, and general signal integrity bottlenecks for 200Gbps data rates. He is an active contributor to IEEE 802.3 and OIF Common Electrical I/O projects. Brandon received the PhD degree in electrical engineering from the University of South Carolina under Dr. Paul G. Huray.



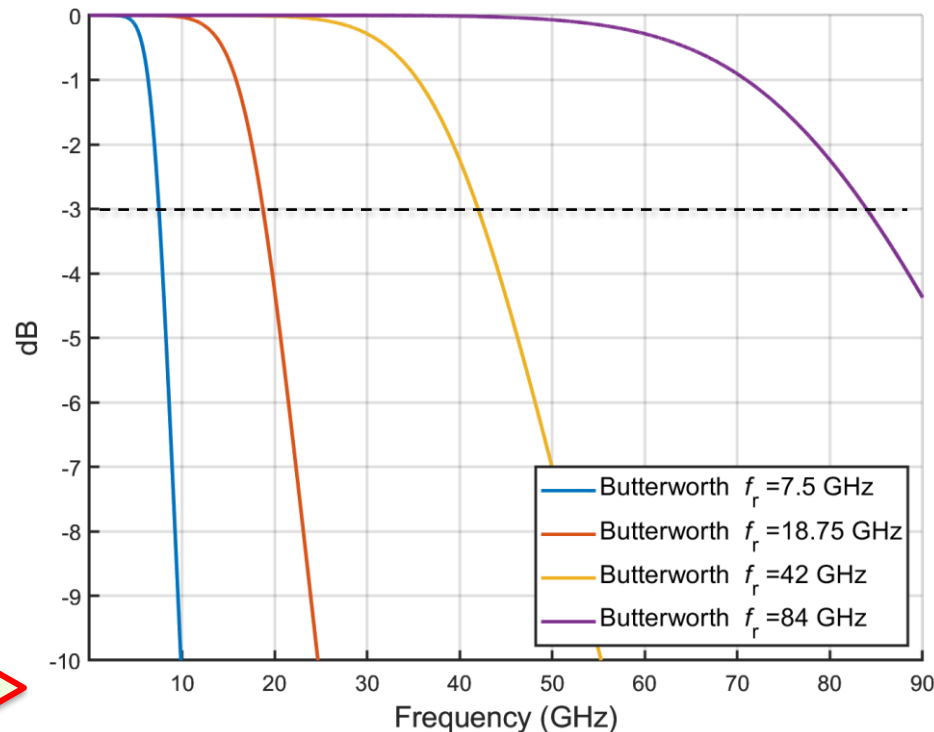
CONTENTS

- **How IEEE Handles Required Rx Bandwidth:**
 - 4th Order Butterworth filter cutoff (f_r)
 - Component verification vs noise verification
- **Modal Theory and RF Cable Assembly Considerations:**
 - Higher Order Mode Cutoff (f_c) as Figure of Merit
 - 1.85 mm (V-band) vs 1.0 mm (W-band)
- **Measurement Case Studies (1.85 mm vs 1.0 mm):**
 - Stripline DUT with Vertical Mount RF Connectors (Top Right)
 - Ganged, cabled RF connectors Passing 224Gb/s PAM4 Signaling (Bottom Right)



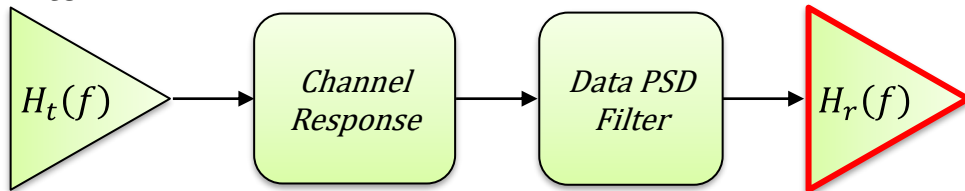
4th Order Butterworth Filter Represents Rx Bandwidth

- Required minimum Rx filter
- Butterworth cutoff (f_r) historically set to $\frac{3}{4}$ of Baud rate i.e. 1.5x fundamental frequency
- f_r relates to max frequency measurement
- 224 Gbps $\rightarrow$ 84 GHz ?



Edge rate
filter

Receiver
filter



Coaxial Higher Order Modes Distort Signal

- Higher order modes → non TEM propagation
- These modes do not exist below cutoff (f_c)
- Coaxial higher order modes are evanescent above cutoff
- First higher order coaxial mode TE₁₁
- Higher order modes with different propagation constant distorts signal → ISI
- Reducing inner ($2a$) and outer diameter ($2b$) or relative permittivity (ϵ_r) increase cutoff

$$J'_n(k_c a) Y'_n(k_c b) = J'_n(k_c b) Y'_n(k_c a)$$

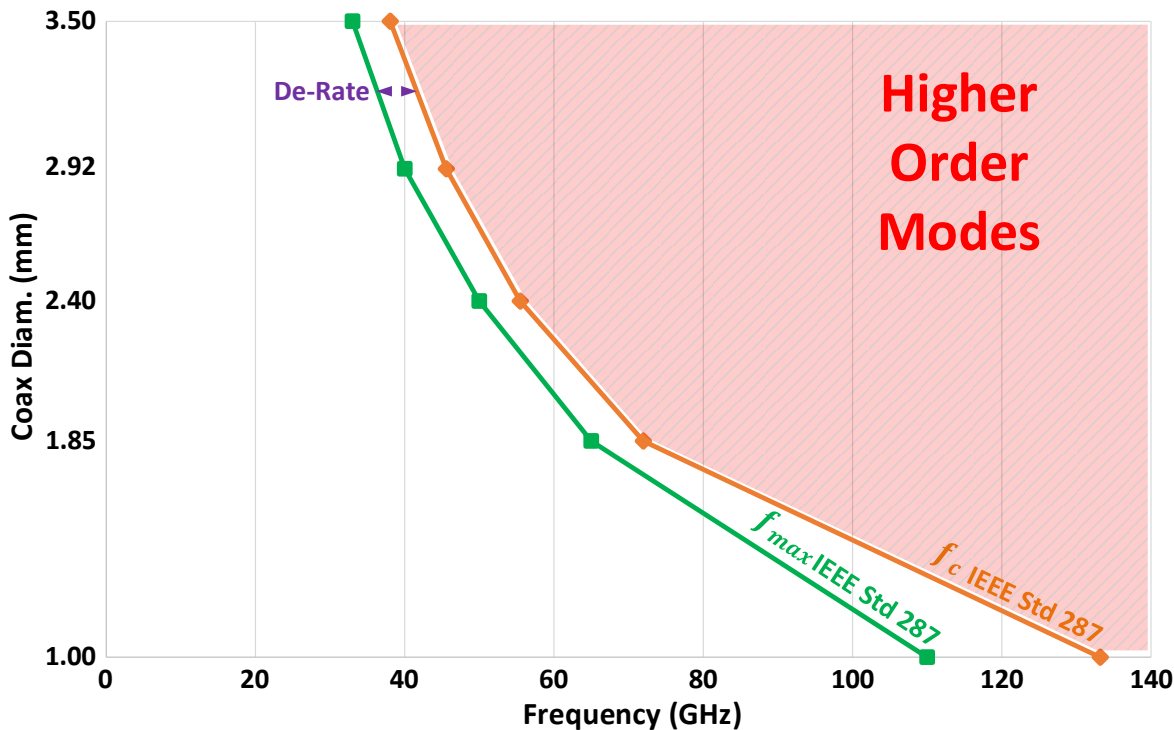
$$f_c = \frac{c_0 k_c}{2\pi \sqrt{\epsilon_r}}$$

$$k_c = \frac{2}{a + b}$$



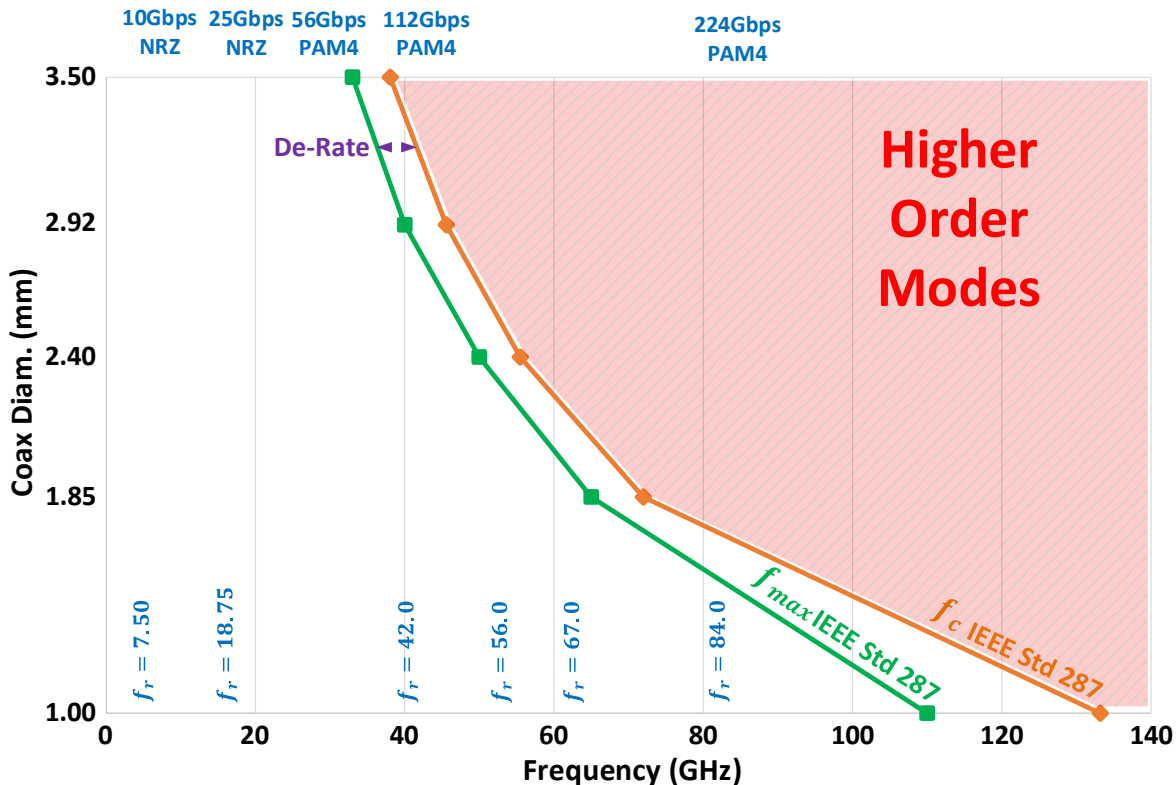
Cutoff Frequency (f_c) Mapped to Usable Bandwidth

- IEEE Std 287 defines expected modal cutoff frequencies (f_c) from mechanical interface specification
- De-rated (f_{max}) for margin and interoperability per coax connector diameter.



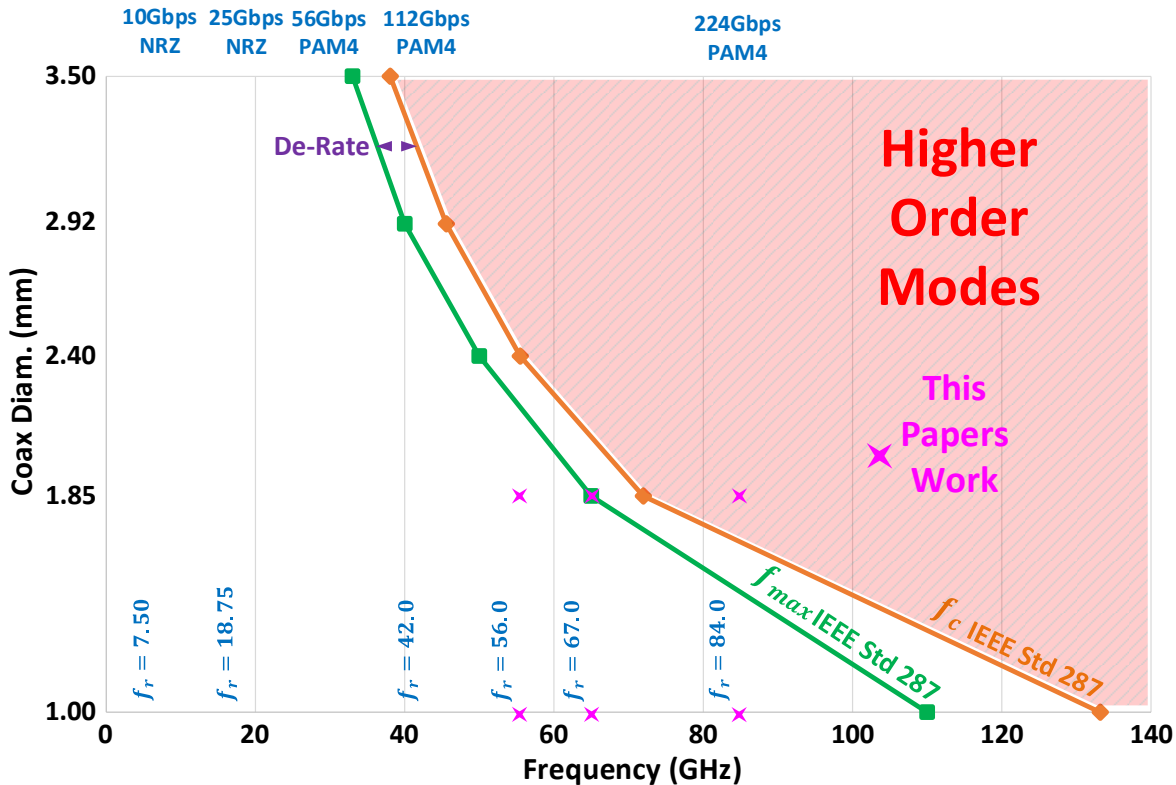
Cutoff Frequency (f_c) Mapped to Usable Bandwidth

- Butterworth filter cutoff (f_r) traditionally 1.5x fundamental frequency for signaling rates 112 Gbps



Cutoff Frequency (f_c) Mapped to Usable Bandwidth

- Compare 224Gbps time domain signal metrics at each of the 6 “starred” data points with a channel emulation board and scope DSP/SerDes Rx emulation.

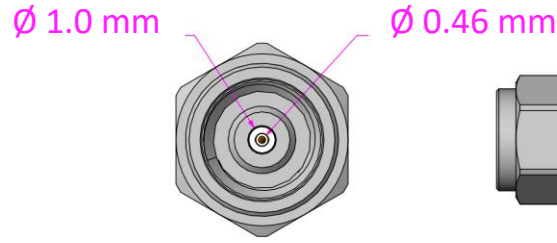


Modal ISI and Implementation Tradeoffs

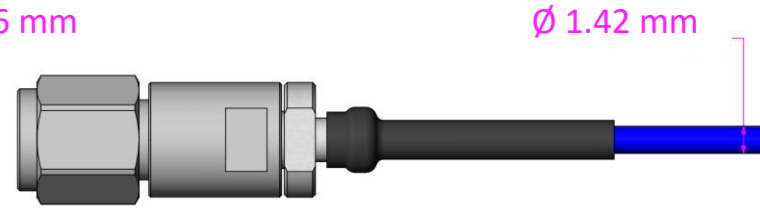


Two Areas of a RF Cable That Can Mode

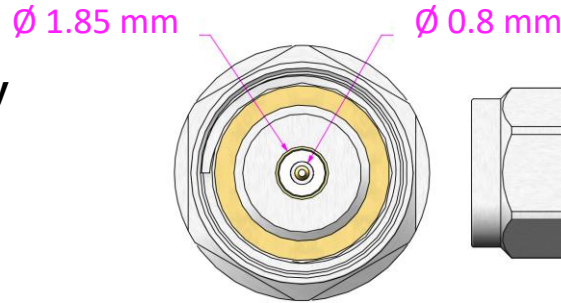
- Airline region of RF connector
- Dielectric support bead complicates
- Cable cross-section limited by air/polymer composites (foaming density)



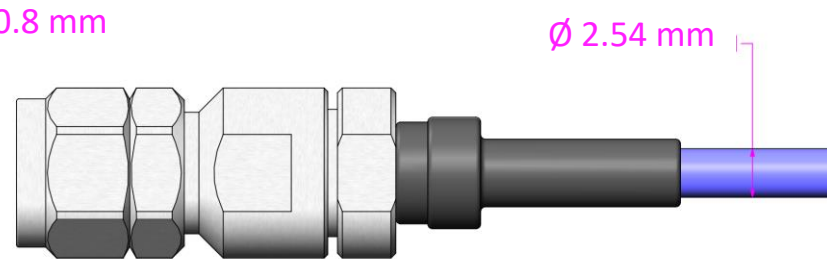
RF047-A-10SP-10SP
SCALE 6:1



RF047-A-10SP-10SP
SCALE 4:1



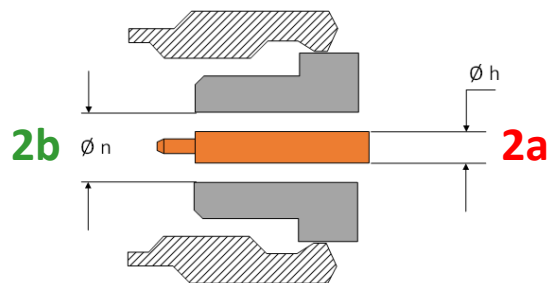
RF086-18SP-18SP
SCALE 6:1



RF086-18SP-18SP
SCALE 4:1

Geometry Shrink to Avoid Modes Introduces Mechanical Challenges

Conductor gap for 1.0 mm shrinks by 46%
Tolerances of 50 um (bore) and 25 um (pin)



2.8X Tighter
Pin Diam. to
Bore Diam.
Tolerance

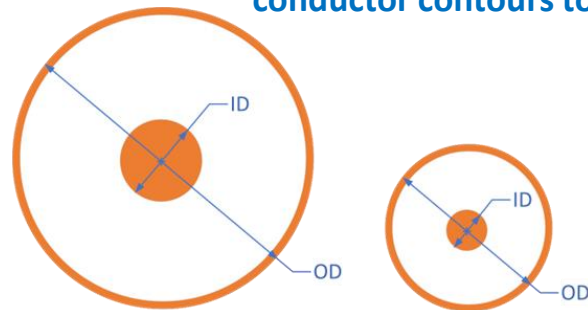
Connector Type	n (mm)		h (mm)	
	nominal	tolerance	nominal	tolerance
1.85	1.85	+/- 0.0127	0.8036	+/- 0.0127
1.0	1.0	+/- 0.0050	0.4340	+/- 0.0025

Cable inner conductor shrinks by 52%

$$k_c = \frac{2}{a + b}$$

$$2\lambda_c = \pi * (2a + 2b)$$

2 Wavelengths must fit
around the sum of the
conductor contours to mode

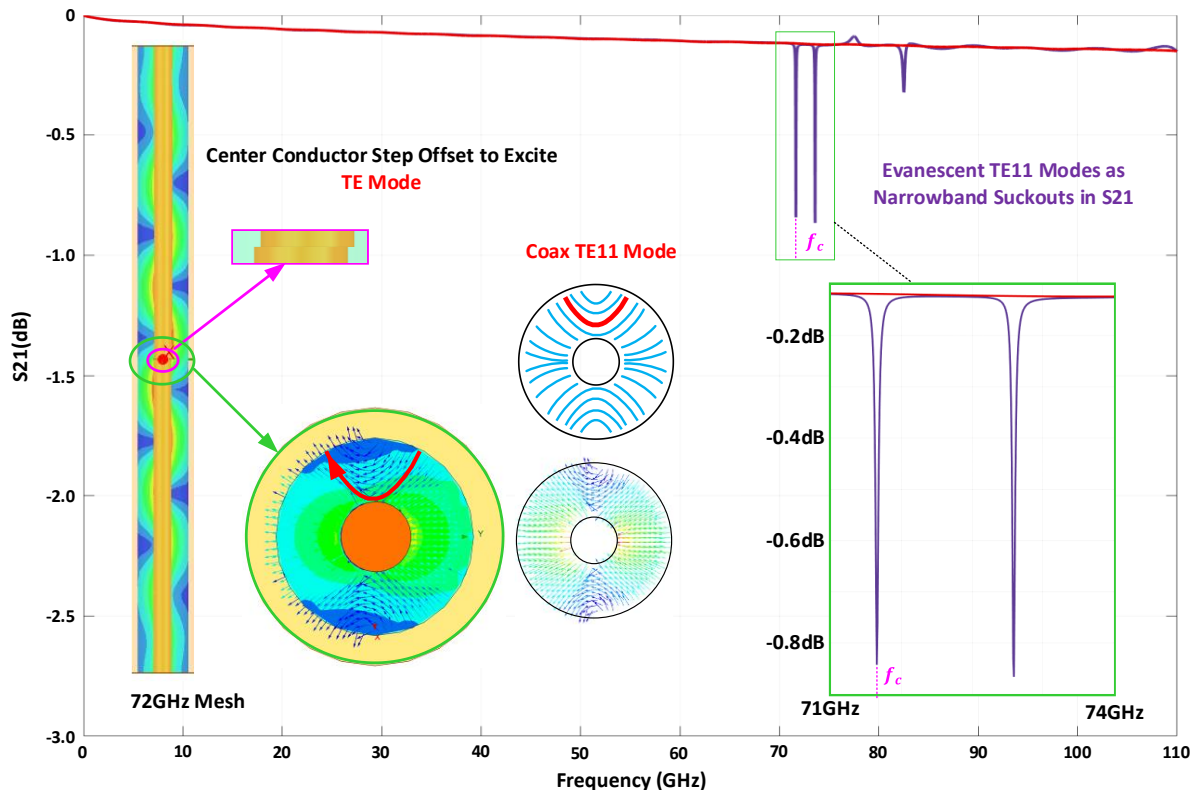


Cable Type	2a ID (in)	2b OD (in)	Cutoff (GHz)
086	0.023	0.086	72
047	0.011	0.047	112



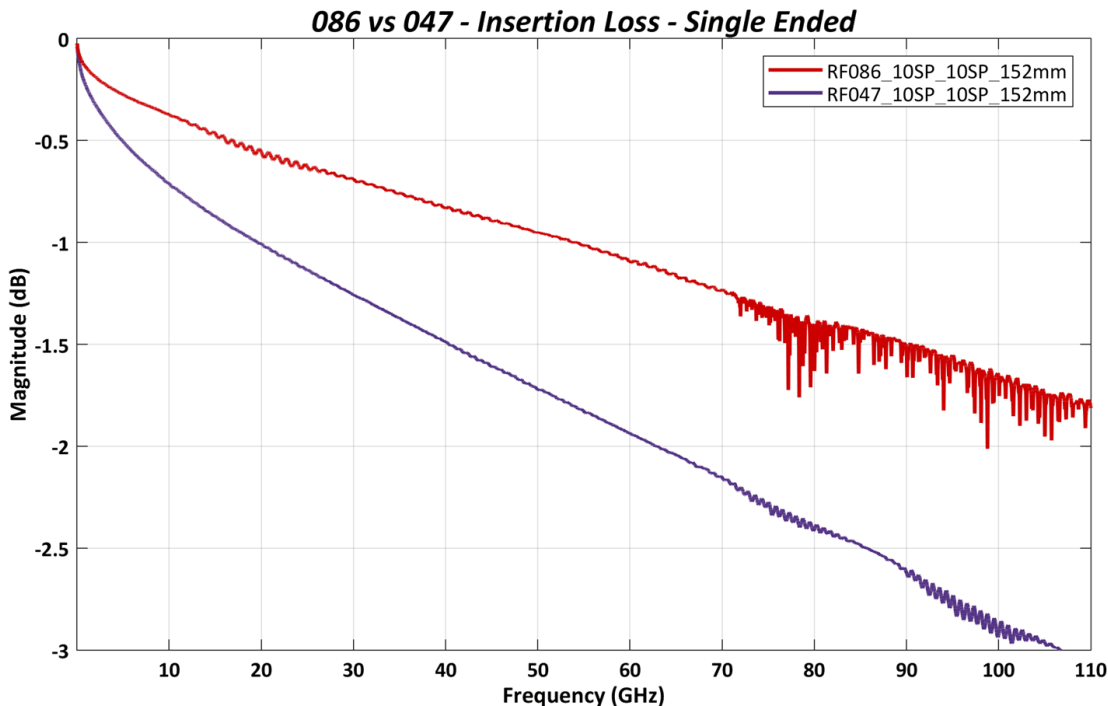
Modal ISI Signature in Cable Geometry

- 3D EM model exploring TE₁₁ coaxial mode excitation through discontinuities
 - Center conductor offset
 - *Verify near-reflectionless loss phenomena*
- Cable cross-section with f_c of 72 GHz
- Evanescent modal excitation present as narrowband suckouts in S₂₁. Energy lost to decay.

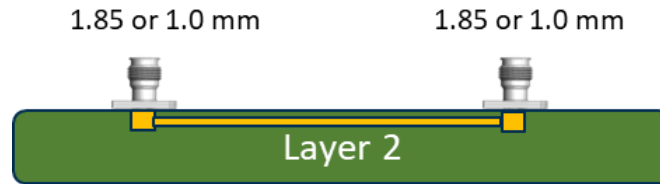


Loss Tradeoff To Avoid Higher Order Modes

- Same length RF cable assembly but with 1.0 mm connectors
- Loss increases due to smaller cable cross-section
- Modal ISI in 086 cable apparent after f_c

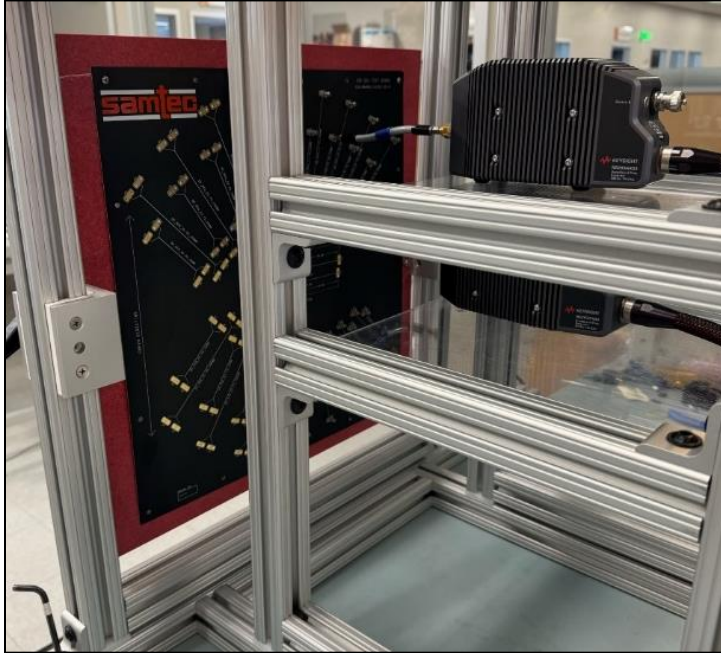


Measurement Case Study No. 1: Compression RF Connectors on a Printed Circuit Board

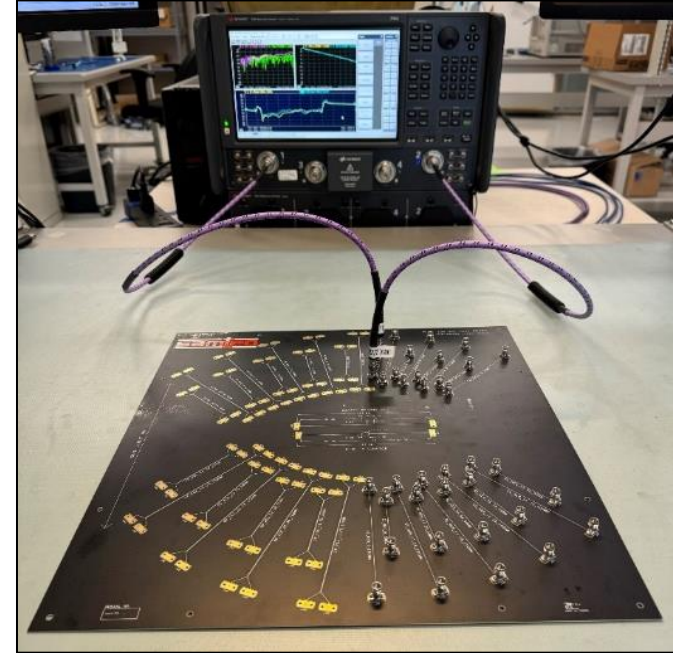


Same Board Swapping Compression Connectors

1.0 mm Test Setup

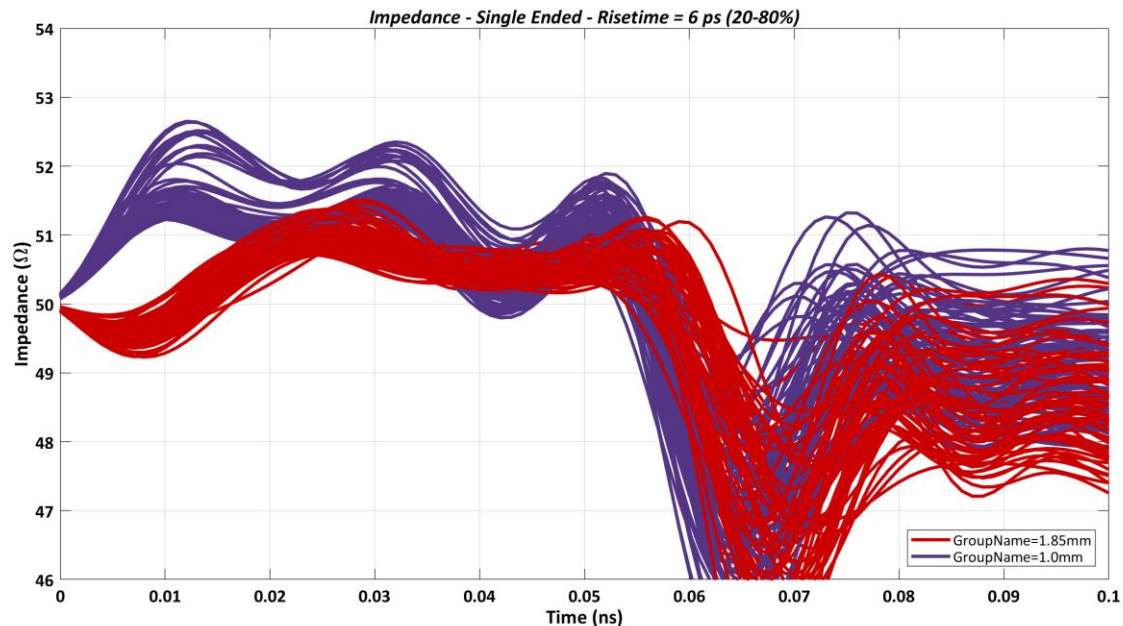


1.85 mm Test Setup

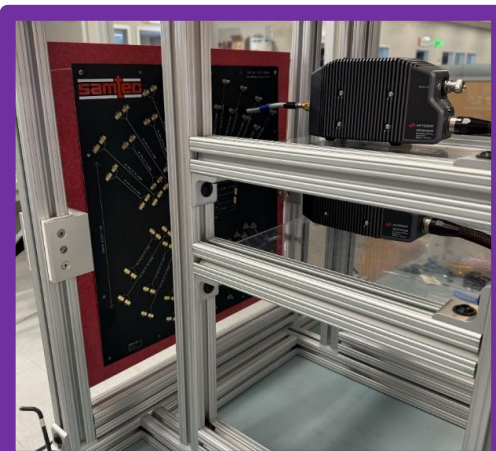


Showing 40 TDR transforms per connector type

More variation of impedance within the connector body for 1.0 mm



1.0 mm

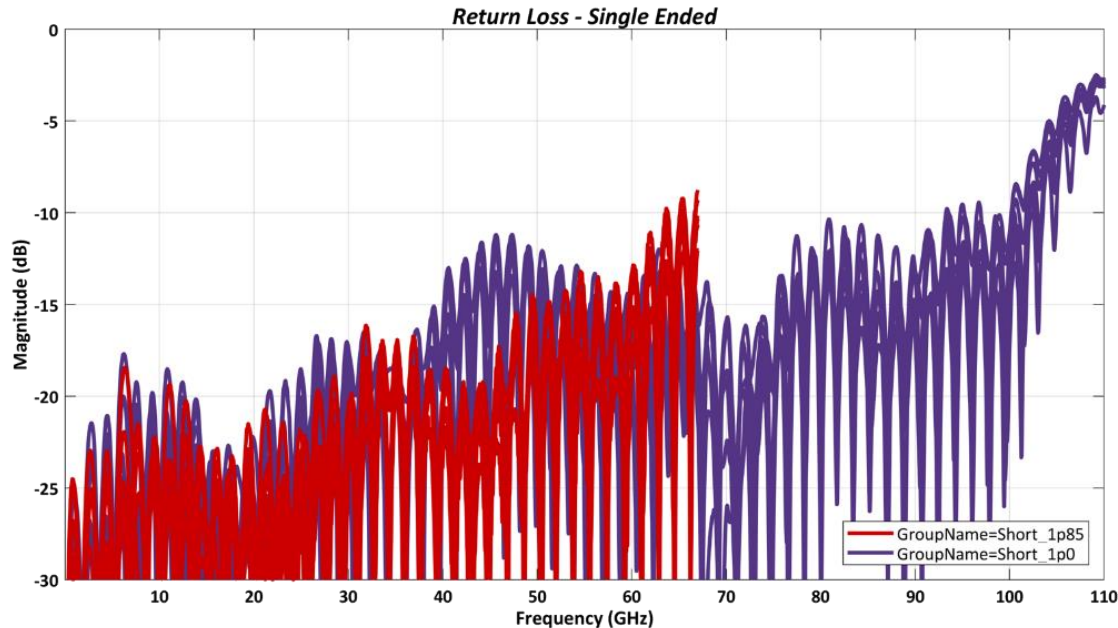


1.85 mm

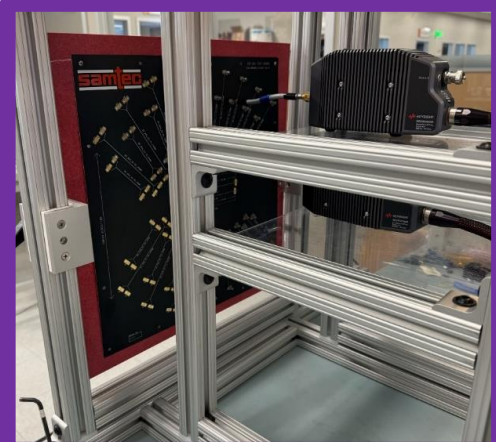


Return Loss was better behaved for 1.85mm

- 1.0 mm more challenging between 40 – 50 GHz
- Critical to have high fidelity models of connector internals



1.0 mm



1.85 mm

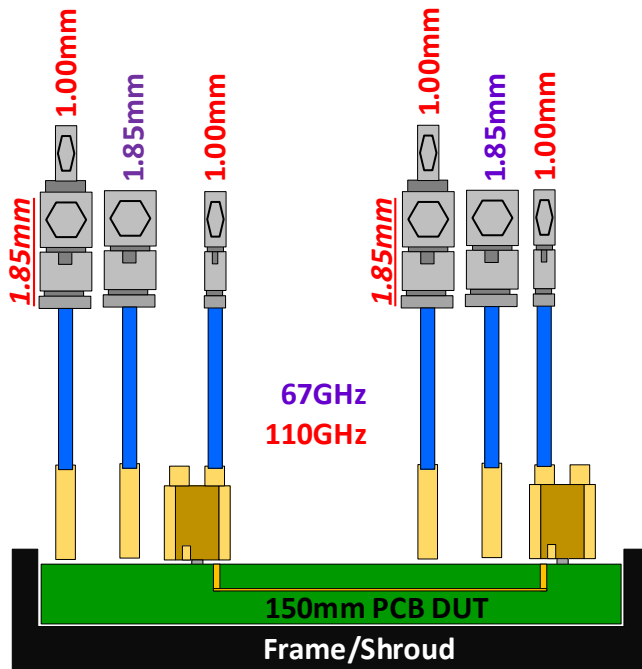


Case Study No. 2: Graduated Loss Platform for Multi Lane Channel Emulation

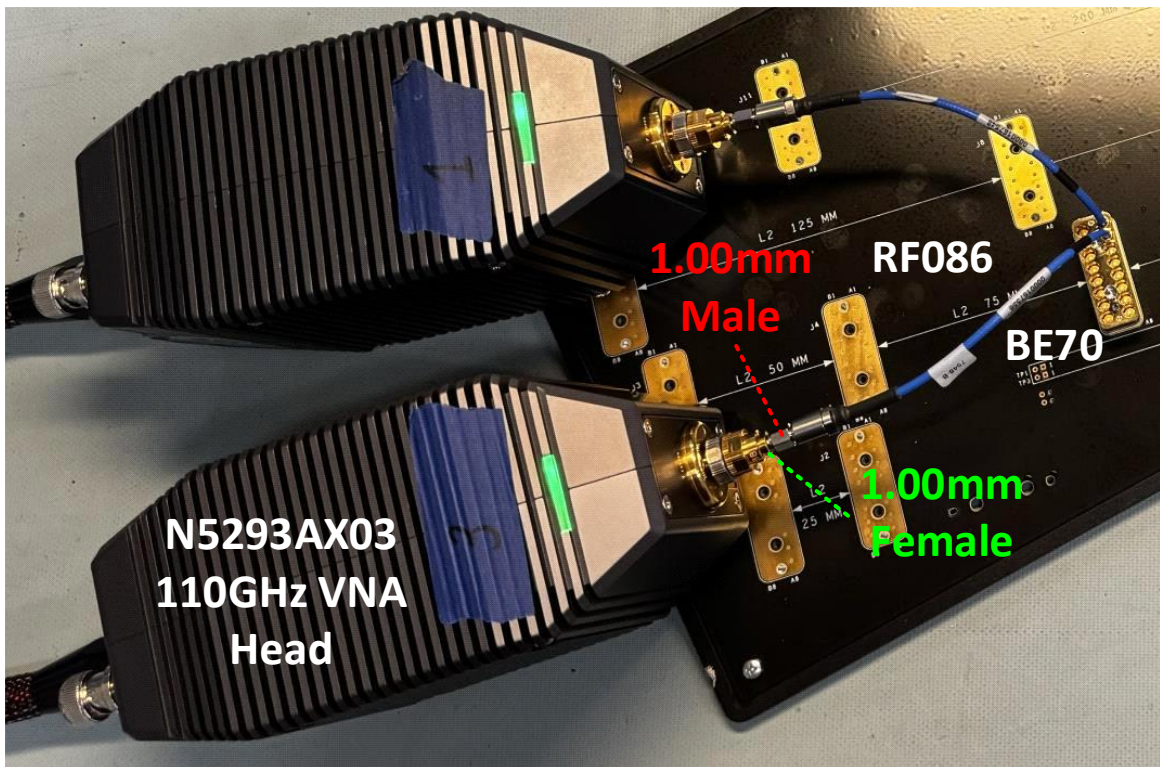
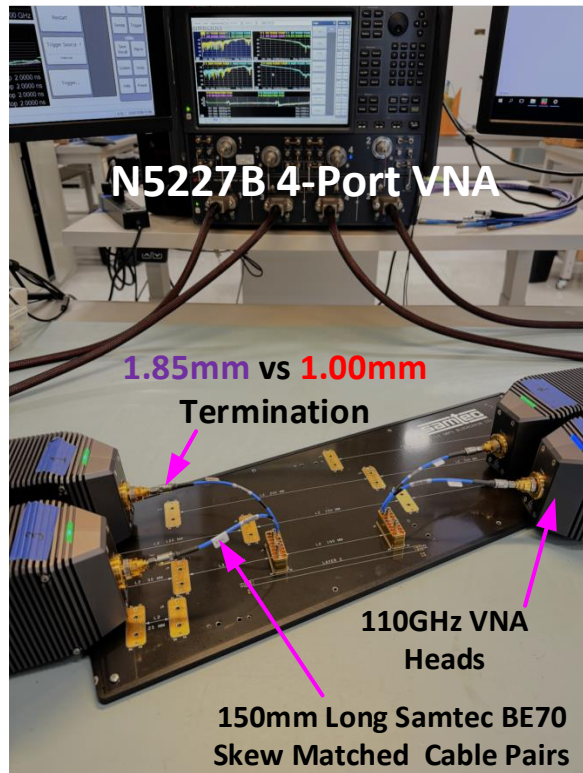


Channel Emulation Featuring 1.0 mm and 1.85 mm

- Cable termination iterations for VNA characterization
- Cable type is 086 for all cases. Why?
 - Constant cable loss
 - Same PCB design
- Test set 1: Characterize with 110 GHz and 67 GHz VNA
- Test set 2: 224 Gb/s PAM4 Signaling

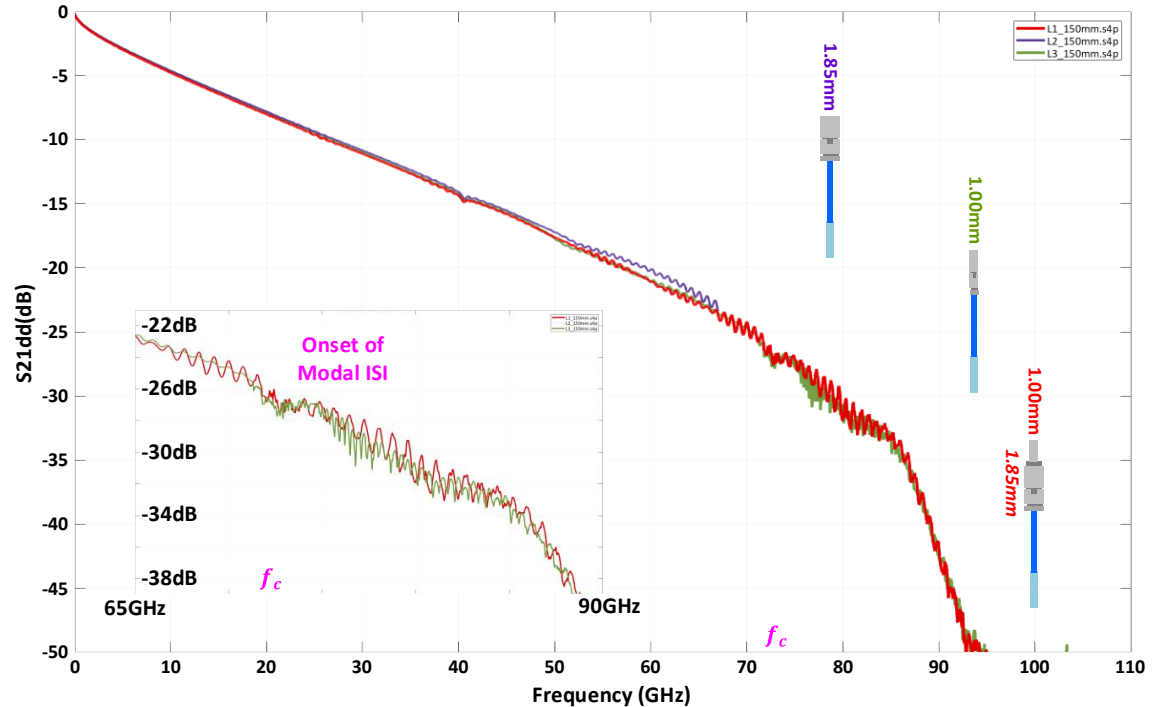


Physical Setup VNA Characterization



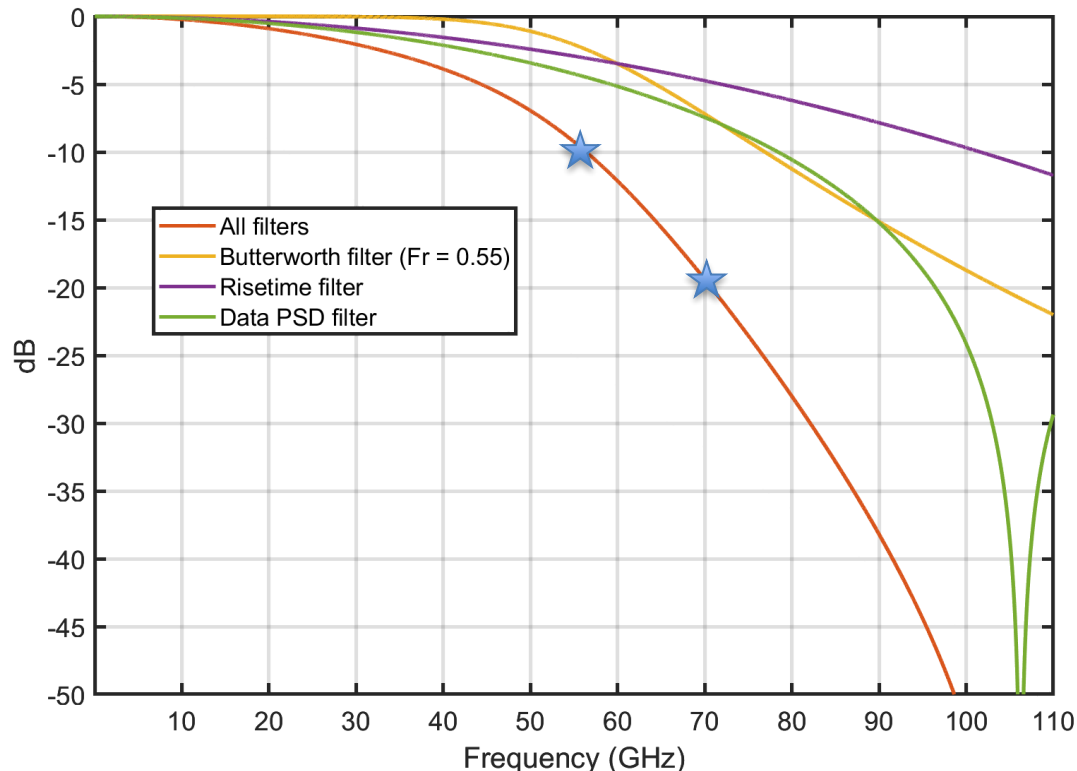
Modal ISI Present in both Cases above f_c Cutoff

- Measured 19 dB Loss @ 56 GHz
- Modal ISI is more apparent in **1.00 mm**
- **1.85 mm** above cutoff is dominated more by impedance mismatch than modal ISI



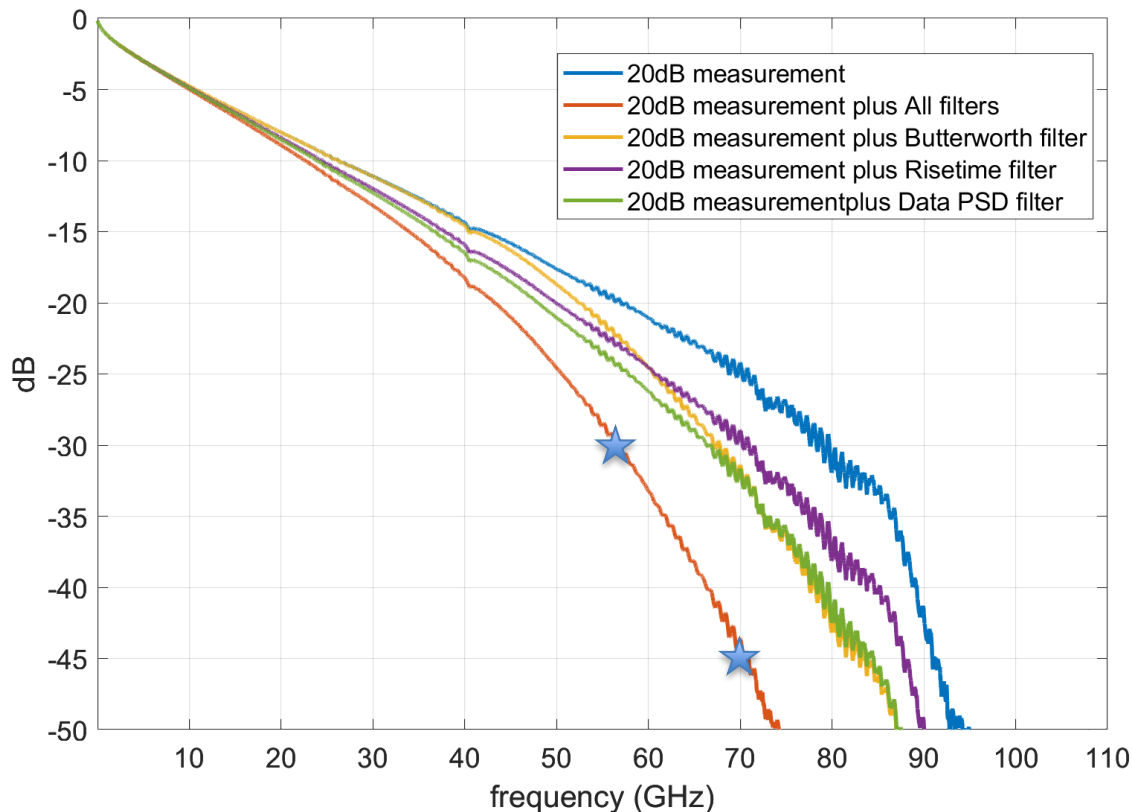
Actual Usage Applies Filters

- Latest trend is Butterworth filter cutoff (f_r) to be 0.55 Baudrate (i.e. 62 GHz) instead of 0.75 Baudrate (i.e. 84 GHz)
- Loss at 56 GHz of ~10 dB after all filters
- Loss at f_c of 70 GHz ~ 20 dB



Effect of Filters on the Channel

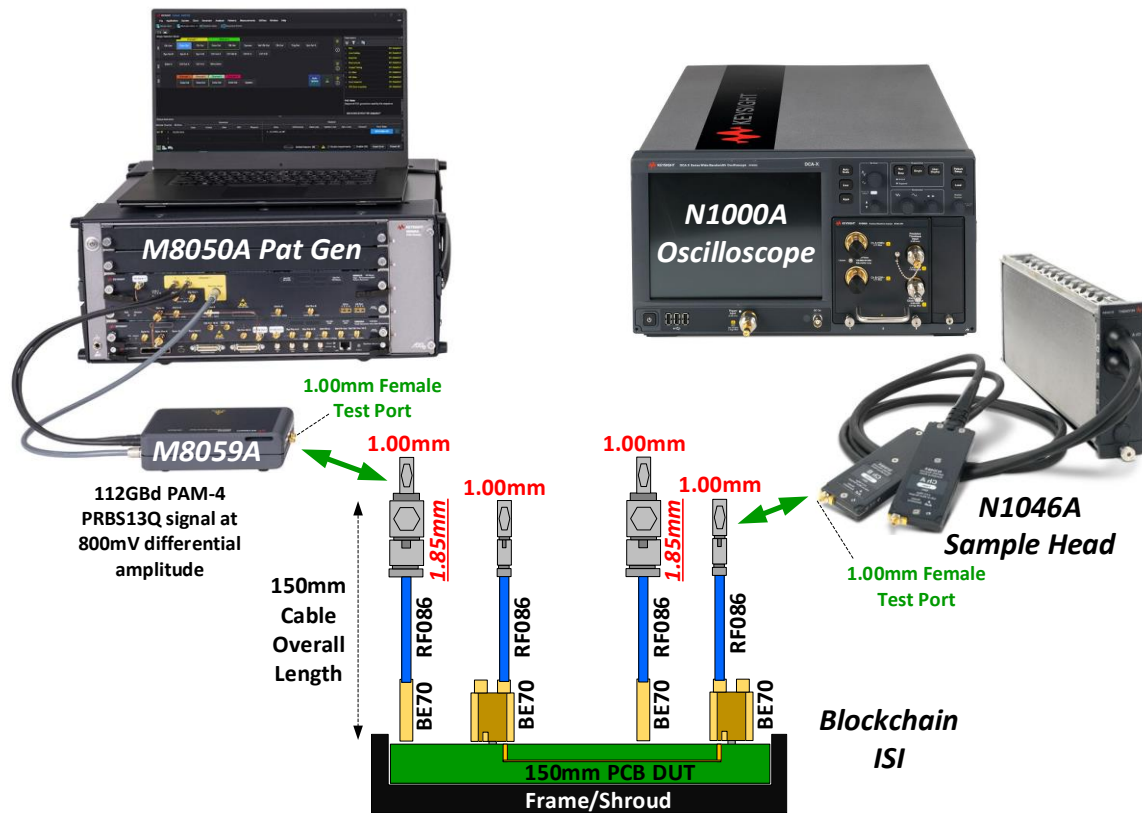
- Consider the measure channel of **1.85 mm** with precision adaptors
- Also consider digital data is a filter
- Modal ISI is heavily filtered when looking in the frequency domain
- ~19 dB Channel Becomes ~30 dB



Test Case 2: 224 Gbps PAM4 Signaling over the Channel Emulation Platform



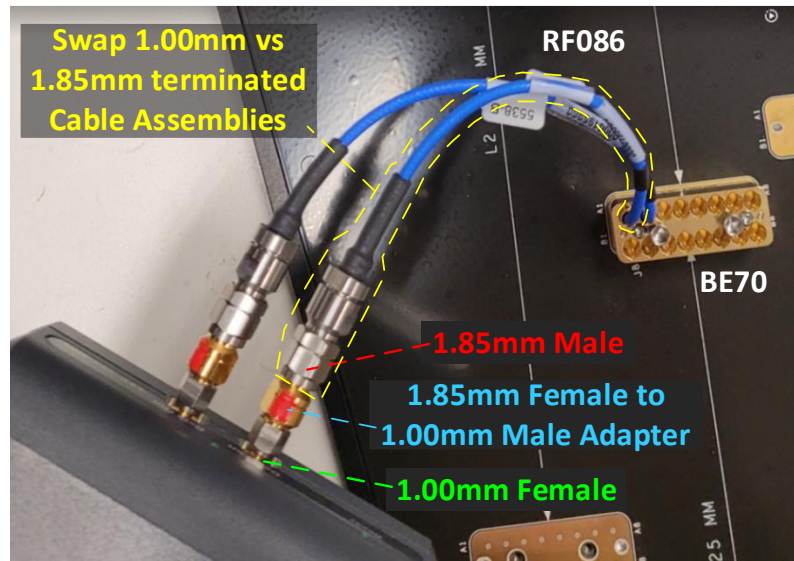
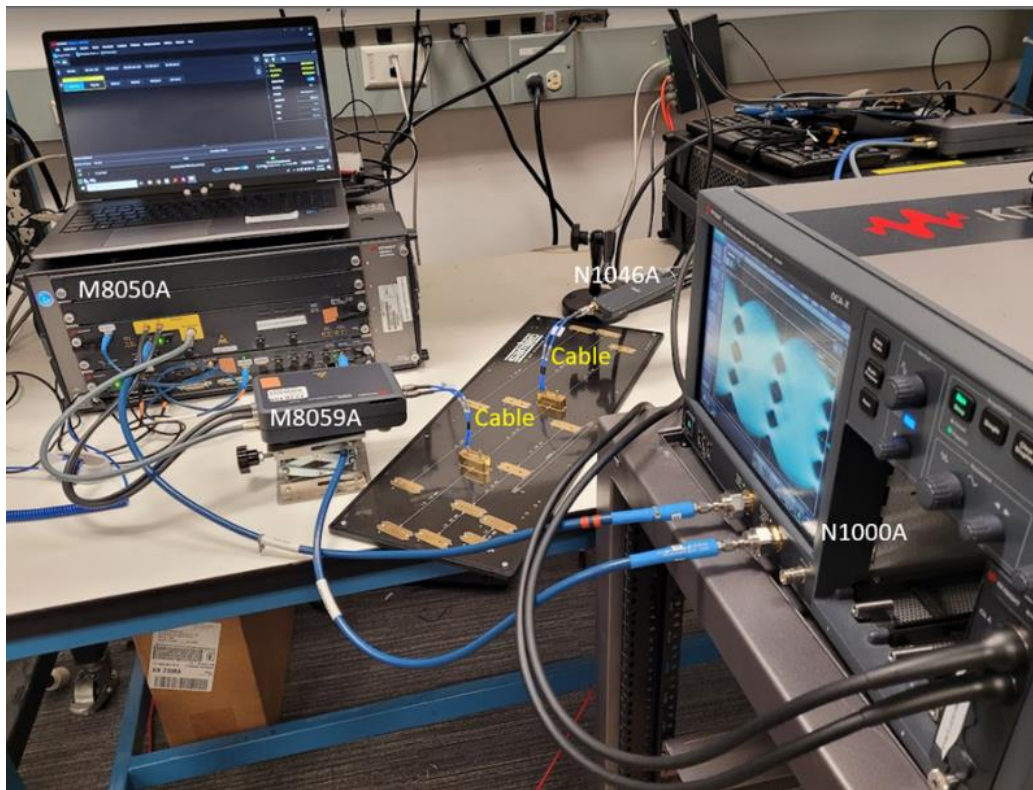
224Gbs PAM-4 HW Setup



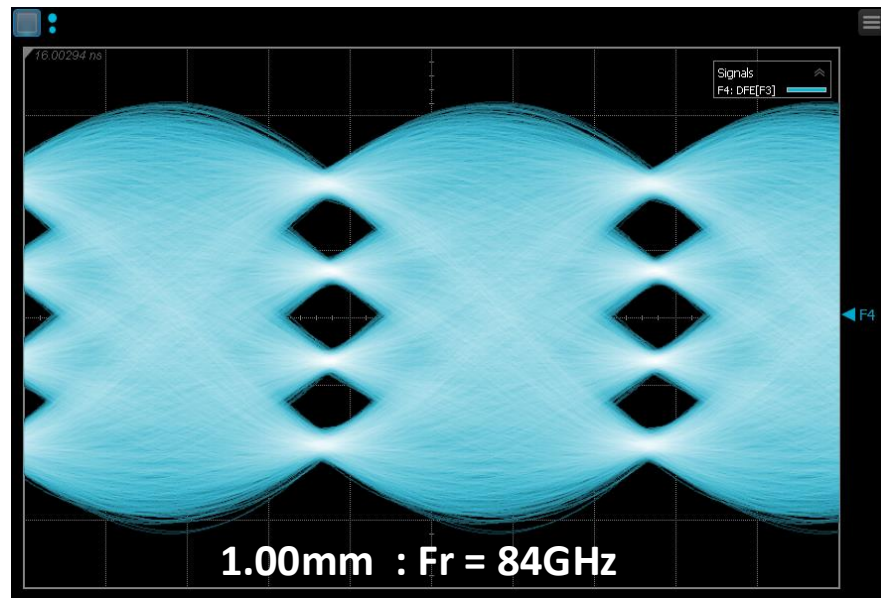
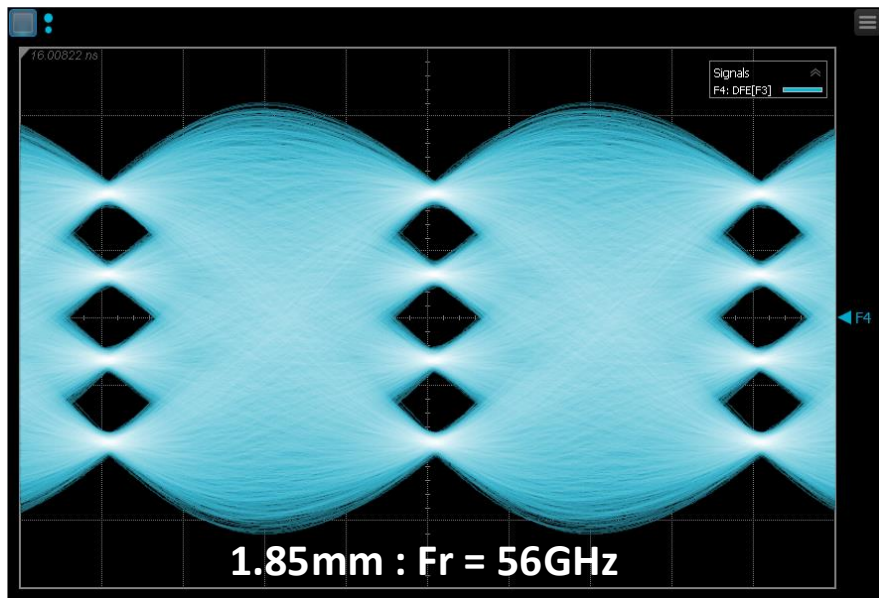
- **N1000A Serdes Rx Emulation**
- **Sweep 4th Order Butterworth Filter**
 - $f_r = 56\text{GHz}$
 - $f_r = 67\text{GHz}$
 - $f_r = 84\text{GHz}$
- **Rx. Eq**
 - CTLE, FFE 6/24, DFE1



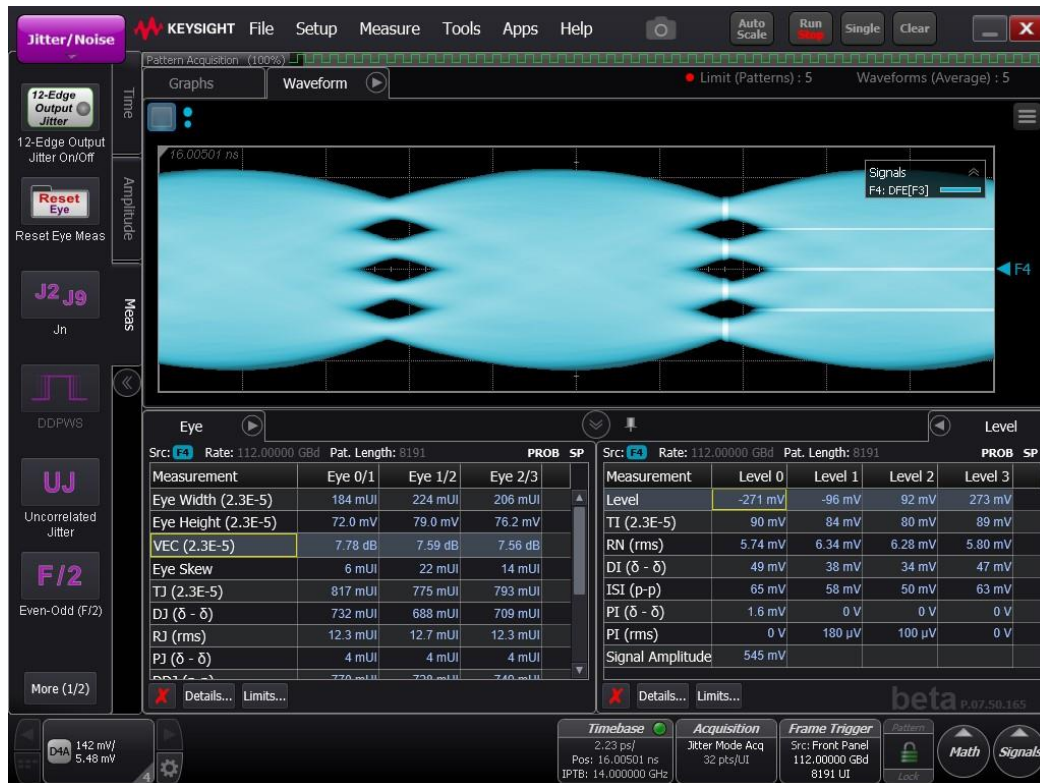
Physical Setup Passing 224Gb/s Signaling



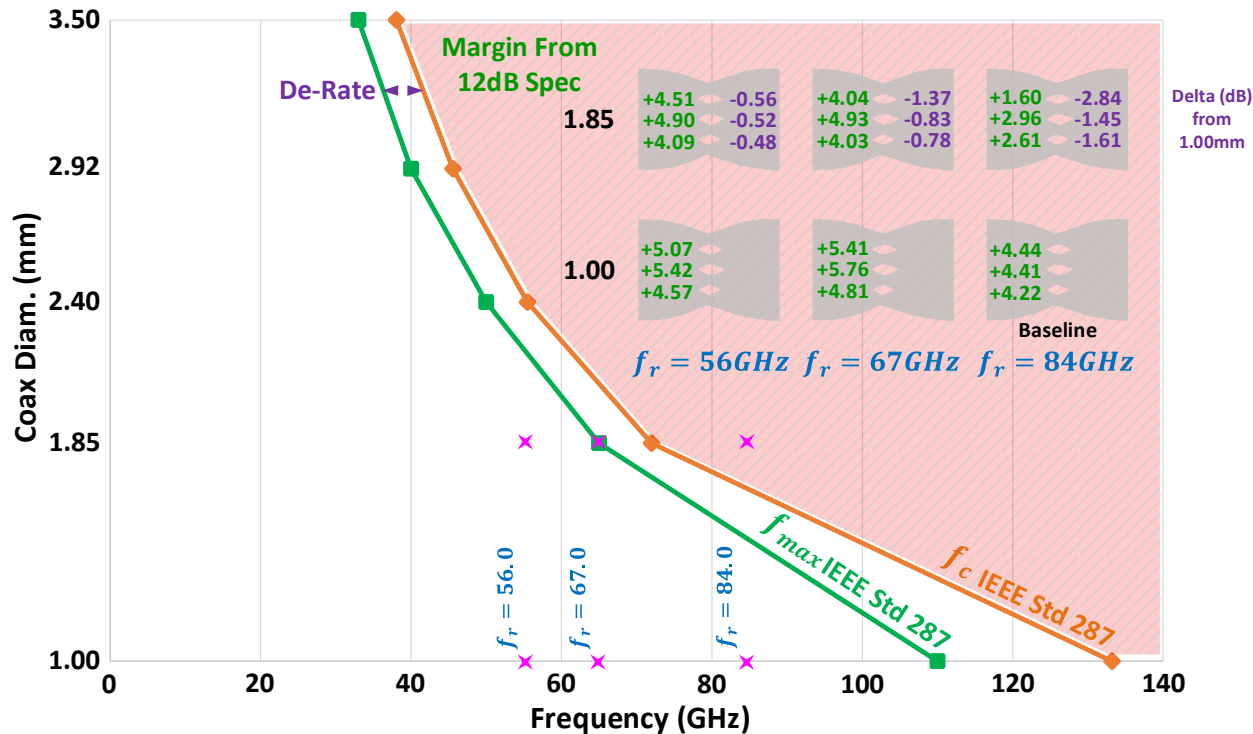
224Gbps PAM-4 Eye Results



Vertical Eye Closure (VEC) Example



VEC Findings



- VEC01, VEC12, VEC23 penalties (dB) from the 1.00mm Baseline
- VEC01, VEC12, VEC23 Margin (dB) to assumed 12dB limit
- 1.0 mm connector best VEC at $f_r = 67\text{ GHz}$
- Lower f_r decreases VEC and Delta (dB) 1.85 mm



Coaxial Connector Feasibility FOM

			$f_{Nyq} = 5 \text{ GHz}$	$f_{Nyq} = 12.5 \text{ GHz}$	$f_{Nyq} = 28 \text{ GHz}$	$f_{Nyq} = 56 \text{ GHz}$		
Conn (mm)	IEEE f_c (GHz)	IEEE f_{max} (GHz)	10G_FOM $\frac{f_{max}}{1.5 * f_{Nyq}}$	25/50G_FOM $\frac{f_{max}}{1.5 * f_{Nyq}}$	100G_FOM $\frac{f_{max}}{1.5 * f_{Nyq}}$	200G_FOM $\frac{f_{max}}{1.5 * f_{Nyq}}$	200G_FOM $\frac{f_{max}}{f_r = 67 \text{ GHz}}$	200G_FOM $\frac{f_{max}}{f_r = 56 \text{ GHz}}$
1.00	133	110	14.67	5.87	2.62	1.31	1.64	1.96
1.85	72	65	8.67	3.47	1.55	0.77	0.97	1.16
2.40	55	50	6.67	2.67	1.19	0.60	0.75	0.89
2.92	46	40	5.33	2.13	0.95	0.48	0.60	0.71
3.50	38	33	4.40	1.76	0.79	0.39	0.49	0.59

It Depends Boundary



Summary of Findings

- Benefits and challenges of 1.0 mm connectors
 - 1.0 mm RF connectors promise higher frequency domain fidelity
 - Test set 2 of 1.0 mm showed quantifiable VEC benefit between 0.5 dB to 1.0 dB
 - Usability constraints, increased costs, and manufacturing penalties are likely
- The choice of 1.85 mm depends
 - 224 Gbps PAM4 copper Standards trending toward an f_r of 0.55 Fb (62 GHz)
 - Either 1.85 mm or 1.0 mm seem OK for receiver, channel compliance, and VEC testing



Thank you!



QUESTIONS?



Resources

- VNA – Test Case 1
 - Keysight N5227B 4-Port PNA Microwave Network Analyzer
 - Keysight N529AX03 110GHz Frequency Extender
 - Keysight N5229A Millimeter-wave Test Set Controller
 - Samtec BE70 High Performance Test Cable Assembly w/Blockchain ISI PCB
- PAM-4 Signaling – Test Case 2
 - Keysight M8050A 120GBd High-Performance BERT w/M8059A Remote Head
 - Keysight N1000A DCA-X Wide Bandwidth Oscilloscope Mainframe
 - Keysight N1046A 100GHz 2-Port Electrical Remote Sampling Head
 - Samtec BE70 High Performance Test Cable Assembly w/Blockchain ISI PCB

